

Signed reflections of an amplitude on symmetric boxes

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2026-09-07

Abstract

Astra #18/#19's rank-2 extension, first unit. For an amplitude η on the symmetric box $(-1, 1]^d$ the coordinate reflections produce the signed symmetrisation $\eta_{\text{sym}}(u) = \sum_{\sigma \in \{\pm 1\}^d} (\prod_i \sigma_i^{h_i}) \eta(\sigma u)$ (`symAmp`, indexed by $\sigma : \text{Fin } d \rightarrow \text{Bool}$), which is continuous when η is, evaluates at the origin to $\eta(0) \prod_i (1 + (-1)^{h_i})$ (`symAmp_zero`), and reduces the symmetric-box dressed integral exactly to the unit-box one:

$$\int_{(-1, 1]^d} \eta(x) x^h e^{-\beta N x^{2k}} dx = \int_{(0, 1]^d} \eta_{\text{sym}}(u) u^h e^{-\beta N u^{2k}} du$$

(`integral_symBox_eq_symAmp`). Zero sorry/axiom.

1 The things formalised

```
def sgn (b : Bool) : Int := if b then -1 else 1
def reflect (h : Fin d → Bool) (u : Fin d → Int) : Fin d → Int := fun i => sgn (h i) * u i
def symAmp (h : Fin d → Bool) (u : Fin d → Int) : Int :=
  (h : Fin d → Bool, (i, sgn (h i) ^ h i) * (reflect h u))
theorem continuous_symAmp (h : Continuous) : Continuous (symAmp h)
theorem symAmp_zero : symAmp h 0 = 0 * (1 + (-1 : Int) ^ h 0)
theorem symAmp_cons (a) (w) : symAmp h (Fin.cons a w)
  = symAmp (Fin.tail h) (fun w' => (Fin.cons a w')) w
  + (-1) ^ h 0 * symAmp (Fin.tail h) (fun w' => (Fin.cons (-a) w')) w
theorem integral_Ioc_symm_add (hg : IntegrableOn g (Ioc (-1) 1)) :
  a in Ioc (-1) 1, g a = a in Ioc 0 1, (g a + g (-a))
theorem integral_symBox_eq_symAmp (h : d h k N, Continuous →
  x in symBox d, x * ((i, x i ^ h i) * exp (-( * N * i, x i ^ (2 * k i))))
  = u in unitBox d, symAmp h u * ((i, u i ^ h i) * exp (-(
  ( * N * i, u i ^ (2 * k i))))
```

2 Mathematics

Induction on the dimension with the Bochner peel on both boxes (unit 188). The inner symmetric integral at fixed first coordinate a is the d -dimensional one with amplitude $\eta(a, \cdot)$ at parameter $N a^{2k_0}$, hence by the induction hypothesis a unit-box integral of the symmetrised section; the peeled coordinate is handled by $\int_{(-1, 1]} g = \int_{(0, 1]} (g(a) + g(-a)) da$, and $(\pm a)^{h_0}$ produces the sign $(-1)^{h_0}$ in front of the reflected section, matching the splitting of the sign sum along `Fin.consEquiv`.

3 Paper-facing meaning

For interior normal coordinates the dressed integral is the unit-box integral of the signed symmetrisation, so the whole continuous-amplitude theory applies with η_{sym} in place of η ; the corner value $\eta_{\text{sym}}(0) = \eta(0) \prod (1 + (-1)^{h_i})$ shows that an odd exponent kills the equal-ratio corner coefficient (next unit), not the integral.

4 Lean notes

- Sum over sign vectors: `Finset.prod_univ_sum` with `Fintype.piFinset_univ` turns $\prod_i \sum_b$ into $\sum_\sigma \prod_i$; `(Fin.consEquiv _).sum_comp` then `Fintype.sum_prod_type` and `Fintype.sum_bool` split the first coordinate.
- `IntervalIntegrable.iff_comp_neg` has an `autoParam`; apply it with named `(f := g) (a := -1) (b := 0)` and `.mp`.
- Using the closure box `piBox d (Ioc (-c) 1)` with $c \in \{0, 1\}$ for both integrability lemmas leaves `Ioc (-0) 1`; rewrite `neg_zero` before `exact`.